

WELLS, ME, USA

WMO: 997707

Lat: 43.320N

Lon: 70.563W

Elev: 20

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.7	(c) 8.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 26.0	(k) 27.0	(l) 23.4	(m) 26.9	(n) 8.1	(o) 310	(p) 0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.2	(c) 84.8	(d) N/A	(e) 80.5	(f) N/A	(g) 77.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 8.2	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 22.4	(b) 19.4	(c) 17.3	(e) -0.5	(f) 91.6	(g) 4.0	(h) 4.0	(i) -3.4	(j) 94.5	(k) -5.7	(l) 96.8	(m) -8.0	(n) 99.0	(o) -10.9	(p) 101.9		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	25.7	28.1	34.4	44.4	54.4	62.0	68.7	67.1	60.7	51.3	41.0	33.0
	DBStd	16.31	10.21	8.44	8.10	6.46	6.89	6.19	5.21	4.43	6.39	6.52	7.75	8.30
	HDD50	2953	755	614	488	190	28	1	0	0	1	62	287	527
	HDD65	6634	1219	1034	949	617	339	131	18	27	160	428	721	991
	CDD50	2102	0	0	5	23	164	360	579	531	323	101	16	0
	CDD65	309	0	0	0	0	10	40	132	93	32	2	0	0
	CDH74	1563	0	0	1	7	131	240	679	343	144	18	0	0
	CDH80	410	0	0	0	0	40	65	196	71	36	2	0	0
(14) Wind	WSAvg	7.6	8.1	8.3	9.1	8.8	7.8	7.1	6.4	6.4	6.4	7.6	7.5	7.5
(15) Precipitation	PrecAvg	46.9	3.4	3.2	4.3	4.0	3.7	4.1	3.8	3.5	3.9	4.8	4.0	4.2
	PrecMax	64.4	6.6	8.5	15.3	9.8	14.6	12.6	7.6	11.3	10.6	14.8	7.8	6.7
	PrecMin	34.6	0.6	1.2	0.9	0.3	0.7	1.1	1.2	0.3	1.4	0.4	1.3	1.4
	PrecStd	8.5	1.2	1.5	2.8	2.2	2.7	2.4	1.9	2.1	2.2	3.3	1.5	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.9	53.1	67.5	72.8	85.9	87.2	91.3	87.1	84.7	76.4	64.1	56.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	47.9	46.2	55.4	64.0	77.0	80.2	85.3	81.3	77.9	67.1	59.4	51.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	44.9	42.2	48.5	58.5	70.1	75.7	80.8	77.7	73.2	63.0	55.6	48.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	41.2	39.6	44.7	54.5	64.3	71.4	77.3	74.6	70.3	60.4	53.1	45.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	14.6	13.5	13.9	13.9	12.6	13.2	13.2	14.1	13.8	13.7	12.3
		MCDBR	17.2	17.2	18.3	21.1	24.8	20.4	19.1	17.7	17.1	16.6	16.9	15.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.291	0.301	0.325	0.357	0.394	0.445	0.461	0.429	0.378	0.336	0.316	0.296
	taud		2.473	2.458	2.464	2.418	2.357	2.245	2.211	2.289	2.409	2.523	2.541	2.535
	Ebn at Noon		270	286	292	288	279	264	258	263	269	270	259	258
	Edh at Noon		23	27	31	35	38	43	44	40	32	26	22	20
(44) All-Sky Solar Radiation	RadAvg		540	816	1160	1466	1638	1758	1851	1683	1350	876	579	435
	RadStd		36	62	73	156	183	147	122	82	77	78	51	47

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+146	+54

Nomenclature: See separate page